

April 5, 2022

Dear Editor,

Thank you very much for sending us the referee's comments to our revised manuscript entitled "Photonic nanolaser with extreme optical field confinement" (Manuscript number: LK17345). We have carefully gone over the comments and suggestions, and have made brief response to all recommendations. Also, in this re-submission, as suggested by the reviewer, we have added an experimental measurement of second-order coherence ($g^{(2)}$), which clearly shows a transition of $g^{(2)}$ to a value very close to 1 at the lasing threshold. Accordingly, to acknowledge the help for $g^{(2)}$ measurement, we have added two students (Jue Gong and Zhanke Zhou) and two additional funds (grants numbers 92150302 and 62175213 from National Natural Science Foundation of China) in the acknowledgement. Finally, a list of changes made in this version is provided at the end of the response.

For your reference, along with clear versions of the revised manuscript and Supplemental Materials, revised manuscript with revision marks is also uploaded.

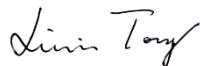
We hope the revised manuscript is now acceptable by PRL.

Thank you very much for your great help!

Sincerely yours,



Xin Guo, Associate Professor



Limin Tong, Professor

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1. *How are the values for the radii obtained from the TEM images? Just by looking at the images, my assessment is that the radii are somewhat larger than the ones the authors arrive at. As the authors mention, I do not think it changes any of the conclusions since the field enhancement is dominated by the small bridge width. However, I think the authors should provide the mathematics of how the radii are obtained. The fitting must have proceeded by optimizing some cost function?*

Response: Thanks for raising this issue. To obtain the radii of the corners, firstly, the TEM image [Fig. R1(a)] is filtered by a gaussian kernel with radius of 0.5 nm for a smoother boundary [Fig. R1(b)] and then a binary image is created from a certain threshold selected as the mean gray value inside the nanowire [Fig. R1(c)]. Then a Hough transformation [1] is applied on the outermost region of the corner to fit the circle and obtain the corresponding radius [Fig. R1(d)]. Because only the outermost region of the corner, which forms the narrowest slit in the CNP, is used for the circle fitting instead of the whole corner, the radii we obtained here seems smaller than the intuition.

Ideally, as the cross section of the wurtzite single-crystal CdSe nanowire is a hexagon, the radius of the corner should be much smaller than we obtained here. However, practically, depending on the growth conditions, the geometry of the corner is usually not that sharp and symmetric, and exhibits a relatively large radius as we obtained here.

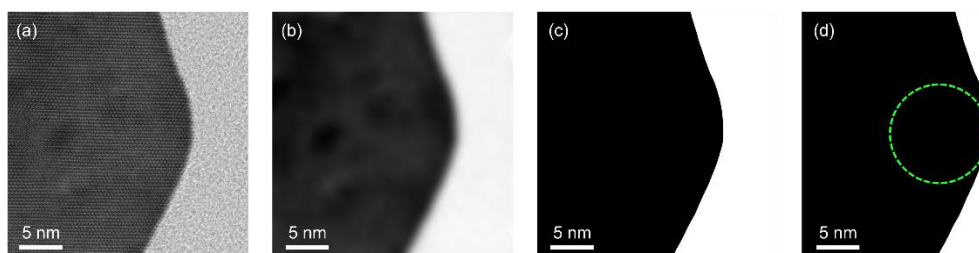


Fig. R1. The process to estimate the radius of the corner. (a) The original TEM image. (b) The image filtered by a gaussian kernel with radius of 0.5 nm. (c) The binary image created from (b). (d) The fitted circle of the outermost corner region by Hough transformation.

2. *The definition of transmissivity seems to be just the transmission coefficient of the end mirror, is that correct? If so, how is it defined for the aperture? I assume it is the power going through the aperture relative to the incident power on the aperture?*

Response: Thanks for raising this issue. For the first question, yes, the definition of transmissivity is just the transmission coefficient of the end mirror. For the second question, it is the power going through the aperture relative to the incident power on the aperture. To maximize the transmissivity, the incident light is assumed to be ideally focused to the optical diffraction limit (i.e., a beam size of $\lambda/2$), which is the smallest beam size we can obtain in free space.

3. *I am still confused about the interpretation of the field confinement in terms of bound electrons. Do I understand it correctly, that the calculation is based on the dielectric polarization given by: $P = \epsilon_0 \chi^{(1)} E$ with E being the electric field. Then, I assume E is calculated using Comsol, which is then used to calculate σ , which is used to argue why the E-field becomes large? In that case, it seems like a circular line of argumentation. The understanding of the field enhancement*

in the references that the authors provide ([8,9]) is that it comes from the boundary conditions of Maxwell's equations. Since I am unfamiliar with the understanding of the enhancement presented by the authors, I think it might be useful to other readers in the same situation if the authors elaborate on this in the supplemental information.

Response: Thanks for raising the point. In fact, our calculation process can be presented by the flow chart in Fig. R2. Based on the Maxwell equation, boundary conditions and the dielectric geometry (the CNP structure), we calculate the electric field distribution $E(x, y)$ using Comsol to obtain the characteristics of the ultratight field confinement, similar to the calculation used in Refs [2,3] (i.e., Refs [8, 9] in the last response). Then, to get a more intuitive explanation, we calculate the polarization $\mathbf{P}$ and surface charge density σ (i.e., $\hat{\mathbf{n}} \cdot \mathbf{P}$) based on the electric field obtained above. Therefore, $\mathbf{P}$ and σ are just the result (not the reason) of the tightly confined field. In addition, the electric field amplitude obtained from the Comsol calculation is a relative value, while the absolute amplitude is determined by many factors including the pump density, quantum efficiency of the material, and the cavity loss.

In the revised Supplemental Materials, we have added the above explanation in Section II.

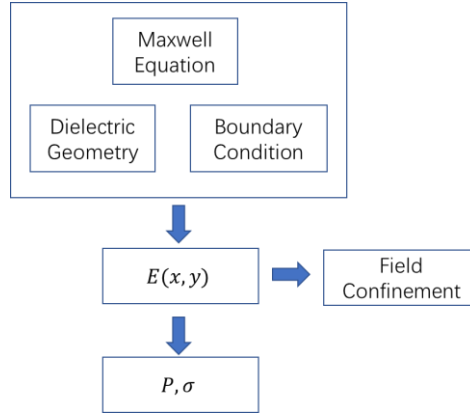


Fig. R2. The flow chart of the calculation process.

4. *The reason I was surprised by the possibility of Γ larger than 1 is that I thought confinement factor means the fraction of the electric field that overlaps with the active material. As far as I know, that is the common way it is used when modeling semiconductor lasers using e.g. rate equations for the photon density and carrier density. Since the authors are using a different definition here, they should perhaps use a different symbol and name. From the author's response, I understand that the reason Γ can be larger than one is that the group velocity is lumped into its definition (which I do not think is the common way to write equations of motion for laser dynamics). But, in the definition of Γ in the supplemental, there is no reference to the group index, rather it is just the material index being used, if I understand correctly. Could the authors comment on this?*

Response: Thanks for raising the point. In the revised manuscript, we have replaced the symbol Γ with Γ_g , where the subscript g denotes the group velocity. In the definition of Γ [Eq. S4 in SM], the group index is included in the Poynting vector P_z , because the Poynting vector describes the energy flux (i.e., z component of $\mathbf{P} = \mathbf{E} \times \mathbf{H}$, where $\mathbf{E}$ and $\mathbf{H}$ are non-zero in all three (x , y and z) directions in a tightly confined nanowire waveguide), which is propagated with the group velocity along the z direction.

In the revised manuscript, we have replaced Γ with Γ_g .

5. *I am not an expert on lasers, but I have understood that there are certain criteria that are often mentioned to support claims of lasing in novel structures. One of them being related to linewidths larger than 0.1nm at- and above threshold. The authors should as a minimum comment on the resolution of the spectrometer used to measure the linewidth and also comment more on the size of the linewidth they measure. The end reflection is very low, so the Q must be quite small (order of 100), so perhaps the authors could also comment on ways of making it bigger (e.g. using FIB to make Bragg mirrors). Ideally, the authors could measure the coherence ($g^{(2)}$) to show a transition to a value of 1 at the lasing threshold. The evidence given from the degree-of-polarization does not seem very convincing, but I also think it should be defined differently and the authors are getting worse numbers due to outliers in the polarization measurements close to 90 degrees (in the insets of Fig. 5a). Shouldn't the DOP be defined in terms of the area enclosed by the blue dots?*

Response: Thank you for raising this issue. The resolution of the spectrometer used to measure the linewidth is 0.035 nm, which is good enough for measuring the 0.5-nm linewidth of the laser peak we obtained here. And exactly as you mentioned, due to the relatively low endface reflection, the Q factor is relatively small. For a Fabry-Perot-type cavity, the Q factor can be calculated as [4]

$$Q = \frac{2\pi L}{\lambda \ln 1/R},$$

where L is the cavity length, λ the lasing wavelength and R the endface reflectivity. With R of 0.05 and L of 20 μm , the Q factor is ~ 60 .

Yes, it's a very nice suggestion to form a Bragg mirror at the end of the nanowire, which can enhance the endface reflection and the Q factor of the cavity, and we'd like to have a try in our further work.

Meanwhile, following your advice, we have measured the $g^{(2)}$ of the endface emission from the sample with 177-nm diameter (sample #3 in section XIII in SM) with the pump density well below (0.6 MW/cm²), close to (1.0 MW/cm²) and above (1.2 MW/cm²) the lasing threshold (~ 1.1 MW/cm²). As shown in Fig. R3, two modes are observed and their horizontal-dominated polarization-dependent-intensity distribution indicates that both modes are TE₀-like modes. When the pump density is increased from well below (0.6 MW/cm²) to approach the lasing threshold (1.0 MW/cm²), a clear thermal bunching of the photons can be observed with $g^{(2)}(0)$ larger than 1; when the pump density is further increased to exceed the threshold (1.2 MW/cm²), the $g^{(2)}(0)$ value decreases to 1, confirming the lasing action.

The definition of the degree of polarization (DOP) in our manuscript is followed by the standard definition of the DOP [5], which is suitable for the situation considered here. Thus, we have not defined the DOP in other ways (e.g., in terms of the area enclosed by the blue dots).

Yes, in the degree-of-polarization measurement, we did have a few worse numbers close to 90 degrees [in the insets of Fig. 5(a)], which is a typical case in endface emission of a deep-subwavelength high-index structure such as the CdSe nanowire we used here, as the orientation of the induced dipoles are not always aligned exactly along the surface of the endface due to the full-vector field of the endface emission. Similar behavior can also be found in Fig. R3(a).

In the revised manuscript, we have added the measurement of $g^{(2)}(0)$ in both main text and Supplemental Materials.

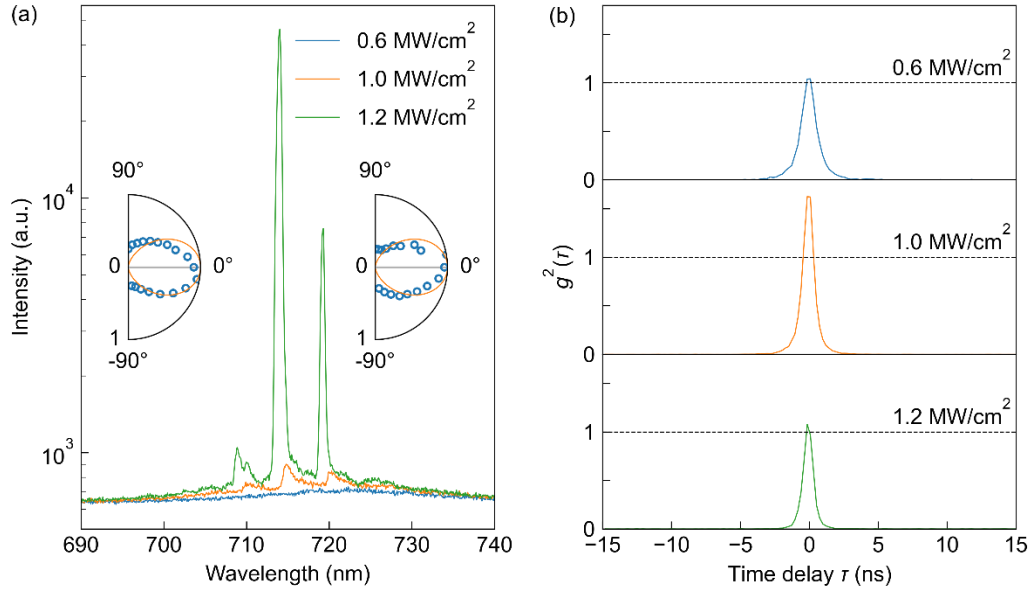


Fig. R3. Emission spectra and $g^{(2)}(\tau)$ curves under three typical pump densities. (a) Emission spectra collected from the endface. Insets: measured (blue circle) and calculated (orange line) polarization-dependent intensity of the two emission peaks at pump density of 1.2 MW/cm². (b) Measured $g^{(2)}(\tau)$ curves well below, close to and above the lasing threshold.

6. *The authors write: “the slit width of this kind of nanolaser may be further scaled down to achieve tighter field confinement”. Since the slit width is already limited by the atomic composition of the surface of the CNP, the authors should elaborate on how they envision a further confinement?*

Response: Thanks for raising the point. Yes, the slit width is limited by the atomic composition of the surface of the CNP. Since the surface roughness of the nanowire can be further improved by improving the growth condition or afterward chemical polishing, it is possible to reduce the slit width to achieve tighter field confinement.

In the revised manuscript, we have mentioned the possible approach to reduce the surface roughness of the nanowire.

7. *I also have a more curiosity-driven question that the authors do not need to respond to, but I would appreciate their opinion/view: For different types of applications, the distance from the end facet of the laser to whichever object is being illuminated must vary (the authors give examples in Fig. S12). Since the mode experiences diffraction after emission from the end facet (as illustrated in Fig. 3), one might expect that tighter confinement inside the laser does not always lead to tighter confinement at the location of the illuminated object because a smaller beam will diffract more. So, perhaps there is an optimum confinement depending on the application? Of course, I appreciate that pushing the limits of confinement is by itself very interesting.*

Response: Thank you for the very insightful comment. Exactly, as we plotted in Fig. 3(b), when the mode emits into free space, the intensity of the peak decreases and the field spreads out. Actually, owing to the very small momentum mismatch between the TE₀-like mode and the free-space light, the divergence and intensity decreasing of this ultra-confined field is much

slower than many other ultra-confined fields generated by nanoplasmonic structures and nano-apertures. Yes, tighter confinement inside the laser does not always lead to tighter confinement at the location of the illuminated object, and there is an optimum confinement depending on the application. Very insightful comment!

References

- [1] R. O. Duda and P. E. Hart, *Use of the Hough Transformation to Detect Lines and Curves in Pictures*, Commun. ACM **15**, 11 (1972).
- [2] S. Hu and S. M. Weiss, *Design of Photonic Crystal Cavities for Extreme Light Concentration*, ACS Photonics **3**, 1647 (2016).
- [3] H. Choi, M. Heuck, and D. Englund, *Self-Similar Nanocavity Design with Ultrasmall Mode Volume for Single-Photon Nonlinearities*, Phys. Rev. Lett. **118**, 223605 (2017).
- [4] *Resonator Optics*, in *Fundamentals of Photonics* (John Wiley & Sons, Ltd, 1991), pp. 310–341.
- [5] M. Born, E. Wolf, A. B. Bhatia, P. C. Clemmow, D. Gabor, A. R. Stokes, A. M. Taylor, P. A. Wayman, and W. L. Wilcock, *Principles of Optics*, 7th ed. (Cambridge, 1999).

List of changes

Main text manuscript:

1. A new affiliation “**Jiaxing Key Laboratory of Photonic Sensing & Intelligent Imaging, Jiaxing 314000, China**” is added for authors Xin Guo and Pan Wang.
2. The symbol of confinement factor is changed from “ Γ ” to “ Γ_g ”.
3. The last sentence of the 2nd paragraph is deleted.
4. In the 17th line of the 4th paragraph, the words “**To be more precise, around the edge of the nanowire, we assume**” are changed to “**with**”, and the words “**from the nanowire (i.e., $n = 2.6$ for CdSe) to the surrounding medium (i.e., $n = 1.0$ for air, see also**” are deleted.
5. In the 5th line of the 5th paragraph, the words “**can be**” are changed to “**is**”.
6. In the last 3rd line of the 5th paragraph, the words “**a photonic mode shown here**” are changed to “**this photonic mode**”.
7. In the 1st line of the 6th paragraph, the words “**It is worth to mention that**” are changed to “**Meanwhile**”.
8. In the end of the 8th paragraph, the sentences “**In addition, calculated quality factor (Q -factor) of the TE₀-like mode of a CNP with 20- μ m length and 170-nm nanowire diameter is ~ 60 . Compared with Q -factors reported in other semiconductor nanowires with larger diameters [24], this Q -factor is relatively lower due to the lower endface reflectivity of a thinner nanowire, which could be enhanced by forming Bragg gratings at one side of the CNP.**” are added.
9. In the last line of the 9th paragraph, the words “**between the photon momentum and spatial confinement**” are deleted.
10. In the 15th line of the 10th paragraph, the words “**by using a plasmonic field, as the required feature size of the gratings should be on the same level as that of the aperture (i. e., 1 nm)**” are changed to “**in this way**”.
11. In the 7th line of the 11th paragraph, the words “**of the bandgap transition**” are deleted.
12. In the 3rd line of the 13th paragraph, the words “**the slit between the two opposite vertex edges of the two nanowires is equivalent to a 1-nm-width slit**” are changed to “**the slit width between the two opposite nanowire vertex edges is about 1 nm**”.
13. In the 4th line of the 14th paragraph, the word “**Since**” is added before “**The measured polarization**”.
14. In the 6th line of the 14th paragraph, the words “**, indicating the observed lasing mode is TE-like mode. Since the nanowire diameter**” are changed to “**and**”.
15. In the 10th line of the paragraph, the words “**that the TE₀-like mode has a lower threshold when $d < 209$ nm and $w = 1$ nm**” are deleted.
16. In the 12th line of the 14th paragraph, the words “**dependence of**” are changed to “**pump-density-dependent**”.
17. In the 13th line of the 14th paragraph, the words “**on pump density**” are deleted.
18. In the 18th line of the 14th paragraph, the words “**(measured with a resolution of 0.035 nm)**” are added after the words “**5 nm to 0.5 nm**”.
19. In the 18th line of the 14th paragraph, the words “**To investigate the evolution of the polarization-dependent intensity of the peak,**” are deleted.

20. In the end of the 14th paragraph, the sentence “In addition, the lasing action of TE₀-like mode has also been confirmed by measuring the degree of second-order coherence [Fig. S18].” is added.
21. In the 1st line of the 15th paragraph, the words “the field confinement on sub-1-nm scale” are changed to “a sub-1-nm field confinement”.
22. In the last 4th line of the 15th paragraph, the word “CdSe” is deleted.
23. In the 2nd line of the 16th paragraph, the words “which can support a lasing field” are deleted.
24. In the 11th line of the 16th paragraph, the words “by extremely localized excitation” are deleted.
25. In the 13th line of the 16th paragraph, the words “using optical gradient force” are deleted.
26. In the 14th line of the 16th paragraph, the words “of field enhancement” are deleted.
27. In the 15th line of the 16th paragraph, the words “of this kind of nanolaser” are deleted.
28. In the 16th line of the 16th paragraph, the words “by using nanowires with lower surface roughness” are added after the words “be further scaled down”.
29. In the acknowledgement, two grant numbers 92150302 and 62175213 are added for the fund National Natural Science Foundation of China.
30. In the end of the acknowledgement, the sentence “The authors thank Yixiao Gao at Ningbo University for helpful discussion.” is changed to “The authors thank Yixiao Gao for helpful discussion, and Jue Gong and Zhanke Zhou for help in lasing characterization.”.
31. In the 4th line of the caption of the Fig. 1, the words “supported by” are changed to “in”.
32. In the 6th line of the caption of the Fig. 1, the words “of the field” are deleted.
33. In the 8th line of the caption of the Fig. 1, the word “view” is deleted.
34. In the 6th line of the caption of the Fig. 2, the words “in a CNP” are deleted.
35. In the 2nd line of the caption of the Fig. 3, the word “nanolaser” is deleted.
36. In the 1st line of the caption of the Fig. 3, the word “coherent” is deleted.
37. In the 3rd line of the caption of the Fig. 3, the word “nano-aperture” is changed to “aperture”.
38. In the last 4th line of the caption of the Fig. 3, the word “1-nm-diameter” is changed to “1-nm”.
39. In the last 3rd line of the caption of the Fig. 3, the words “in each case” are deleted.
40. In the 2nd line of the caption of the Fig. 3, the word “output” is deleted.
41. In the 2nd line of the caption of the Fig. 4, the word “typical” is deleted.
42. In the 4th line of the caption of the Fig. 4, the word “Close-up” is deleted.
43. In the 5th line of the caption of the Fig. 4, the word “a” is changed to “the”.
44. In the 6th line of the caption of the Fig. 4, the words “formed in a CNP” are deleted.
45. In the 7th line of the caption of the Fig. 4, the word “illustration” is deleted.
46. In the 8th line of the caption of the Fig. 4, the words “optical pumping and” and “of the CNP” are deleted.
47. In the 11th and 13th lines of the caption of the Fig. 4, the words “with pump density” are deleted.
48. In the 14th line of the caption of the Fig. 4, the word “boundary” is changed to “edge”.
49. In the 1st line of the caption of the Fig. 5, the word “lasing” is changed to “photoluminescence”.
50. In the 4th line of the caption of the Fig. 5, the words “The distribution” are changed to “Distribution”.
51. In the 5th line of the caption of the Fig. 5, the words “nanowire diameter” are deleted.

Supplemental Materials:

52. In the beginning of the Section II, the sentences “Based on Maxwell’s equations, boundary conditions and the dielectric geometry (the CNP structure), we calculate the electric field distribution $E(x, y)$ using a finite element method (COMSOL Multiphysics®), similar to the calculation used in Refs [2,3]. Then, we calculate the polarization P and surface charge density σ (i.e., $\hat{n} \cdot P$) based on the electric field $E(x, y)$ obtained above.” are added.
53. In the Section VII, the symbol “ T ” is changed to “ T_g ”.
54. The title of the Section XIII is changed to “Characterization of emission from CdSe CNP samples under pumping”.
55. In the end of the Section XIII, we add the details of $g^{(2)}$ measurement and Fig. S18.